

Homework — 1.4.1 Data Types

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.4.1 lesson. *SHOW ALL WORKING.*

Part A — This week's topic (~14 marks)

1. Complete the conversion grid. Binary values are 8-bit unsigned. **[3]** (AO2)

Denary	8-bit binary	Hexadecimal
156	(a)	(b)
(c)	11001010	(d)
111	(e)	6F

2. Using 8-bit **two's complement**, calculate **52 – 37** by adding the two's complement of 37. Show the binary working and state clearly what happens to the final carry. **[4]** (AO2)

3. Write **True** or **False** for each statement about shifts and masks. If a statement is false, write a correction in the final column. **[2]** (AO1/AO2)

Statement	True / False	Correction if false
Applying an arithmetic right shift of two places to 01001100 gives 00010011, dividing the value by 4.		
ANDing a value with the mask 00001111 sets its four leftmost bits to 1.		
Shifting a binary value one place to the left multiplies it by 2 (provided no bit is lost).		
XORing a bit with 1 leaves that bit unchanged.		

4. Write the letter of the correct definition in the **Match** column next to each term. Each letter is used exactly once. **[3]** (AO1)

Term	Match	Letter	Definition
Character set		A	The error that occurs when the result of a calculation is too large to represent in the number of bits available
ASCII		B	The part of a floating point number that holds its significant digits, determining the precision
Unicode		C	The complete set of characters a computer can represent, each given a unique binary code
Overflow		D	A character set with enough codes for the symbols of all the world's major languages
Mantissa		E	The part of a floating point number that determines where the binary point is placed, giving the range
Exponent		F	A character set that uses 7 bits, giving 128 codes — enough for the English alphabet, digits and common symbols

5. A games developer stores a player's score, which is always a whole number, in a **real (floating point)** variable. Explain, with reference to how integers and real numbers are represented, **one** reason why an **integer** data type is the better choice. [2] (AO1/AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.1.1) State the role of the **MAR** (Memory Address Register) in the fetch–decode–execute cycle. [1] (AO1)

7. (1.2.1) Describe how **virtual memory** allows a computer to run a program that is larger than the installed RAM. [3] (AO1)

8. (1.3.1) State **one** difference between **lossless** and **lossy** compression, and give a suitable use for each. [2] (AO1)
